

Economic Analysis of Text and other Non-Standard Data

ISEG Summer School 2026

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9. Generative AI

Outline

Training Large Language Models

BERT

GPT

Gen AI for YOUR Research

Intro: Generative AI for Text

- ▶ **Type of generative AI that produces text**
- ▶ AI systems trained to predict the next word given preceding text
- ▶ Typically fine-tuned to follow human instructions and generate responses aligned with human preferences
- ▶ Based on deep neural networks with billions of parameters
- ▶ Built on transformer models (with attention mechanisms, which endogenously assign varying degrees of importance to different words)

Step 1: Pre-Training

- ▶ Calculate conditional probability distribution over words given the preceding words, based on its training data (**Next Token Prediction**)
- ▶ **Self-Supervised Learning**: Model is fed text fragments; parameters are adjusted to predict continuation
- ▶ Terabytes of data (Wikipedia, scientific articles, books, etc.)
- ▶ Neural nets learn language structure:
 - ▶ Syntactic structures
 - ▶ Relationships between words and concepts they represent
 - ▶ Context of sentences and word interactions
 - ▶ Relationships between different sentences

Step 2: Instruction Fine-Tuning

- ▶ Improves model to follow human instructions
- ▶ Example: Pretrained model might predict a continuation of “What is your name?” as “And how old are you?”; this is not what we want
- ▶ Instruction fine-tuning trains the model to respond appropriately
- ▶ **Supervised learning:** Feeding the model millions of examples for responding to thousands of different instructions
- ▶ Used for tasks like summarization, answering questions, brainstorming, etc.

Step 3: Reinforcement Learning

- ▶ Improves model by incorporating human feedback
- ▶ Human raters provide feedback on different responses
- ▶ Helps align model with human preferences, particularly in areas hard to define via instruction fine-tuning
- ▶ Example: Penalizing hateful responses
- ▶ Noisy process:
 - ▶ Partly why LLMs sound authoritative even when hallucinating

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Autoencoding Language Models

- ▶ **Main example: BERT**

- ▶ e.g. **BERT = “Bidirectional Encoder Representations from Transformers”**
- ▶ pretrained by dropping/shuffling input tokens and trying to reconstruct the original sequence.
- ▶ usually build bidirectional representations and get access to the full sequence.
- ▶ can be fine-tuned and achieve great results on many tasks, e.g. text classification.

BERT I/II

- ▶ Task: Masked language modeling:
 - ▶ 15% of words masked
 - ▶ if masked: replace with [MASK] 80% of the time, a random token 10% of the time, and left unchanged 10% of the time.
 - ▶ model has to predict the original word.
- ▶ Architecture:
 - ▶ Architecture:
 - ▶ A stack of Transformer blocks, each containing a multi-head self-attention layer and an MLP.
 - ▶ BERT-Base has 12 blocks, a hidden dimension of 768, and 12 attention heads: approximately 110M parameters.
 - ▶ BERT-Large has 24 blocks, a hidden dimension of 1,024, and 16 attention heads: approximately 340M parameters.

BERT II/II

- ▶ Unlike GPT, BERT attention observes all tokens in the sequence, reads backwards and forwards (bidirectional).
- ▶ Corpus:
 - ▶ 800M words from English books (modern work, from unpublished authors)
 - ▶ 2.5B words of text from English Wikipedia articles
- ▶ Various extensions:
 - ▶ RoBERTa: improves BERT's pre-training procedure.
 - ▶ DistilBERT: a smaller and faster version of BERT.
 - ▶ VisualBERT: jointly processes text and image-region features.

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Autoregressive Language Models

- ▶ **Main example: GPT**

- ▶ e.g. **GPT** = “**Generative Pre-Trained Transformer**”:
- ▶ pretrained on classic language modeling task: guess the next token having read all the previous ones.
- ▶ during training, attention heads only view previous tokens, not subsequent tokens.
- ▶ ideal for text generation.

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 - ▶ trained on the Books corpus (7000+ books)
 - ▶ train on a language modeling task, as well as a multi-task that adds a supervised learning task.

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 - ▶ all articles linked from Reddit with at least 3 upvotes (8 million documents, 40 GB of text)
 - ▶ dispense with supervised learning task, make some other architectural adjustments
 - ▶ make model much bigger

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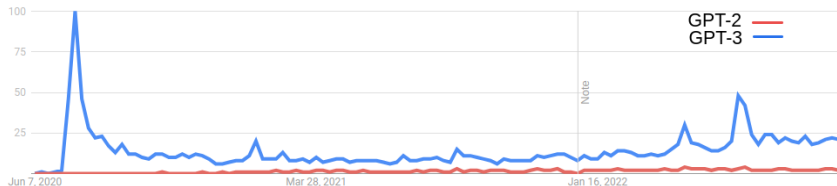
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- ▶ GPT-3 (2020):
 - ▶ use an even bigger corpus (Common Crawl, WebText2, Books1, Books2 and Wikipedia)
 - ▶ make model much, much bigger

OPENAI'S NEW MULTITALENTED AI WRITES, TRANSLATES, AND SLANDERS

A step forward in AI text-generation that also spells trouble

By James Vincent | Feb 14, 2019, 12:00pm EST

Howard, co-founder of Fast.AI agrees. "I've been trying to warn people about this for a while," he says. "We have the technology to totally fill Twitter, email, and the web up with reasonable-sounding, context-appropriate prose, which would drown out all other speech and be impossible to filter."



BUSINESS

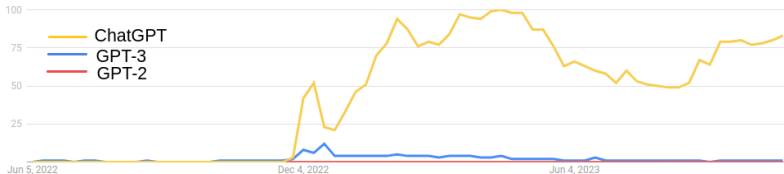
Is ChatGPT the Start of the AI Revolution?

Analysis by The Editors | Bloomberg

December 9, 2022 at 1:39 p.m. EST

The New Chatbots Could Change the World. Can You Trust Them?

Siri, Google Search, online marketing and your child's homework will never be the same. Then there's the misinformation problem.



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- ▶ GPT-4 (2023):
 - ▶ Multimodal model: can take also images as inputs
 - ▶ Trained in two stages:
 1. token prediction (like other GPT models)
 2. **reinforcement learning with human feedback**
 - ▶ Much, much, much bigger model

GPT-5.6 (OpenAI): What matters for researchers (July 2026)

- ▶ **Positioning:** OpenAI flagship for **coding** and **agentic** (tool-using) tasks; released July 2026 as a family (flagship “Sol”; cheaper “Terra”/“Luna” variants).
- ▶ **Context + outputs:** $\approx$ 1M-token context window; up to 128k max output tokens (API).
- ▶ **Modalities:** text + **image input**; text output (no native audio/video).
- ▶ **Reasoning control:** `reasoning.effort` parameter (none/low/medium/high/xhigh/max).
- ▶ **Knowledge cutoff:** Feb 16, 2026 (API documentation).

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Coding agents in the stack: Claude Code vs Codex

Foundation model (Claude / GPT) → **Product surface** (app/API) → **Agent tool** (runs actions)

- ▶ **Claude Code (Anthropic):** agentic coding tool that *reads your codebase, edits files, runs commands*, and integrates with dev tools (terminal/IDE/CI).
- ▶ **Codex (OpenAI):** coding agent in your terminal that can *inspect repos, edit files, and run commands*; included with several ChatGPT plans.
- ▶ **Lecture takeaway:** Claude/GPT are *models*; Claude Code/Codex are *agent products* that add permissions + tool execution (and therefore add security/reproducibility risks).

Choose your GenAI model (updated July 2026)

LLM	Price	Tokens ¹	Data Cutoff	Images	PDFs upload
GPT-5.6 (OpenAI)	free / 8\$ (Go) / 20\$ (Plus) / 200\$ (Pro)	≈1M ctx (API); 128k max output	02/16/2026	yes (image input)	yes
Claude Fable 5 / Sonnet 5 (Anthropic)	free / 20\$ (Pro) / 100\$+ (Max)	1M ctx; 128k max output	01/2026	yes	yes
Gemini 3.1 Pro (Google)	free / 19.99\$ (AI Pro) / 100–200\$ (AI Ultra)	1M input / 64k output	01/2025	yes	yes
DeepSeek-V4 (Pro/Flash)	free (web/app) + API	1M ctx; 384k max output	not stated	indirect	yes (app: file upload + text extraction)
Llama 4 / Qwen 3.5 (open weights)	open weights (self-host)	1M–10M ctx (Llama 4)	08/2024 (Llama 4)	yes (multimodal)	no (needs tooling)

- ▶ Beyond raw model quality, choose based on ecosystem: connectors, coding agents, file tooling (OCR/PDF upload), and admin/security controls.
- ▶ Open-weight frontier is now Meta **Llama 4**, Alibaba **Qwen 3.5** (Apache-2.0), and **DeepSeek-V4**; great for on-prem + cost control, but you must build the surrounding chat-app features (uploads, RAG, UI, monitoring/guardrails).

¹Token calculator: <https://platform.openai.com/tokenizer>

Generative AI for Economic Research

- ▶ Korinek (2025)²:
 - ▶ Provides an overview of how researchers can exploit these technologies

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 - ▶ Discusses long-run implications of AI for economic research

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 - ▶ Discusses long-run implications of AI for economic research
- ▶ Main takeaway on LLMs for research:

“employ AI agents in research is to treat them like a PI would treat a team of research assistants: they require careful planning of what is to be done, oversight during execution, and detailed vetting of the final results.”

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“employ AI agents in research is to treat them like a PI would treat a team of research assistants: they require careful planning of what is to be done, oversight during execution, and detailed vetting of the final results.”
- ▶ Good practices:
 - ▶ Provide context (e.g., specify point of view, audience, etc.)
 - ▶ Iterate
 - ▶ Be patient

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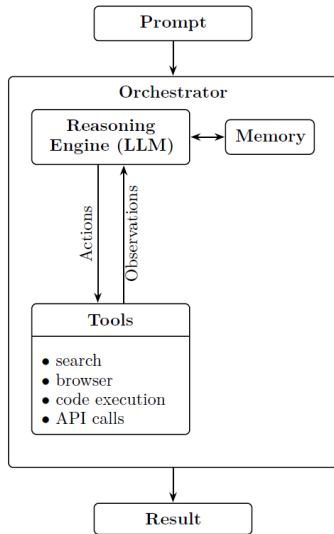
Three paradigms: Chatbots → Reasoners → Agents

- ▶ **Traditional LLMs** (broadly available since Nov 2022): fast, fluent, single-pass responses.
- ▶ **Reasoning models** (introduced Sep 2024, e.g. OpenAI o1): step-by-step analysis for harder math/logic/coding.
- ▶ **Agentic chatbots** (consumer tools from early 2025, e.g. OpenAI Operator): language + reasoning + *autonomous use of external tools*.
- ▶ By 2026: reasoning is a standard toggle in flagship models (GPT-5.x, Gemini 3.x, Claude Opus/Fable), which auto-select modes and “think longer” when needed.

What is an AI agent? (Working definition)

- ▶ An AI agent combines:
 - ▶ a large language model (LLM),
 - ▶ planning (decide steps),
 - ▶ memory (track intermediate results),
 - ▶ tool access (web search, code execution, APIs, files).
- ▶ Unlike a standard chatbot, an agent can:
 - ▶ decompose a question into subtasks,
 - ▶ gather information from multiple sources,
 - ▶ execute code and iterate based on intermediate findings,
 - ▶ synthesize results into coherent narratives.

AI Agent Architecture



Comparison of LLM model types

Table 1
Primary Usefulness of LLM Model Types for Research Tasks

Research Category	Traditional LLMs	Reasoning Models	Agentic Chatbots
Ideation & Feedback	Good for initial brainstorming.	Best for structured feedback and identifying logical flaws.	Actively scans literature for novelty and grounding.
Writing	Excellent for drafting, summarizing, and rephrasing existing text.	Ensures logical flow in complex arguments.	Incorporates real-time web information.
Background Research	<i>Shortcoming: No live web access.</i>	Synthesizes info from provided texts.	Best for live web searches and up-to-date literature reviews.
Coding	Generates basic code snippets.	Excellent for writing and debugging complex algorithms.	Executes code, tests hypotheses, and interacts with data files.
Data Analysis	<i>Shortcoming: Cannot execute code.</i>	Helps interpret data and suggest approaches.	Best for end-to-end analysis: data cleaning, coding, visualization.
Math	<i>Shortcoming: Unreliable for complex math.</i>	Solves multi-step problems and formal proofs at PhD level.	Can leverage external computational tools to ensure accuracy.
Promoting Research	Drafts initial promotional content.	Tailors summaries for specific audiences.	Automates creating summaries and posts for multiple platforms.

Note: The most useful model type in each research category is bolded.

Deep Research agents: multi-agent literature review at scale

- ▶ Core architecture:
 - ▶ **Lead/orchestrator** plans the strategy and decomposes the question into subtasks.
 - ▶ **Specialized subagents** investigate threads in parallel (each with its own context/tools).
 - ▶ **Synthesis** integrates findings into a cited report.
- ▶ Strength: compresses time-intensive information gathering + initial synthesis dramatically.
- ▶ Important caveats:
 - ▶ Often compiles existing knowledge (not genuinely novel insight),
 - ▶ Can struggle to identify the most impactful frontier papers,
 - ▶ Steering depth/breadth is constrained by provider cost limits and usage caps.

Coding agents & “vibe coding”

- ▶ Coding agents: terminal-based assistants that can generate, modify, test, and debug multi-file projects.
- ▶ “Vibe coding”: building software end-to-end from natural-language prompts.
- ▶ Example workflow (econometrics):
 - ▶ build a small UI for CSV upload,
 - ▶ select variables for OLS,
 - ▶ compute regression stats (SEs, t -stats, p -values, R^2),
 - ▶ plot scatter + regression line,
 - ▶ iterate quickly via follow-up prompts (fix missing values, fix distributions, etc.).
- ▶ Broader use: even if you hand-code, these tools are strong for code review and bug-finding.

Under the hood: the basic agent loop

- ▶ A simple agent can be structured as:
 1. **Think:** decide what information/tool calls are needed,
 2. **Act:** call tools/APIs (e.g., data retrieval),
 3. **Observe:** parse/validate returned outputs,
 4. **Respond:** produce a grounded natural-language answer.
- ▶ Key design idea: make intermediate outputs explicit so they can be checked, logged, and debugged.
- ▶ Economists: this pattern supports reproducible pipelines (data access → analysis → reporting).

Limitations

Practical limitations (why human oversight remains essential):

- ▶ hallucinations (incl. bogus citations/statistics),
- ▶ error propagation / multi-step cascades,
- ▶ brittleness to prompt changes (reproducibility issues),
- ▶ prompt injection risks when tools/web are involved,
- ▶ not yet reliable as independent economic reasoners.

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- ▶ Have GPT validate pairs of clauses:
“Which of these sentences from a union collective bargaining agreement is more likely to be interpreted as an entitlement, benefit, or amenity for workers? Answer ‘Definitely 1’, ‘Probably 1’, ‘Probably 2’, ‘Definitely 2’, or ‘Neither’. 1. [sentence 1]. 2. [sentence 2].

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- ▶ Compute the probability that one clause is more pro-worker than all others
- ▶ Validate previous classification

Pairwise Comparisons – Arold et al. (2025)

Table 4: LLM Validation of Worker Rights as Favoring Workers

Clause Type	Clause Frequency (%)	Pro-Worker Frequency (%)
Worker Right	22.9	79.7
Worker Permission	8.4	67.3
Union Right	2.1	65.7
Firm Obligation	24.7	60.7
Manager Right	0.2	58.4
Worker Obligation	20.9	53.1
Manager Obligation	1.7	44.5
Union Permission	2.0	43.2
Worker Prohibition	3.1	41.6
Manager Prohibition	0.1	38.8
Firm Prohibition	1.5	36.5
Firm Permission	3.4	35.5
Manager Permission	0.4	34.3
Union Obligation	7.0	31.3
Union Prohibition	0.6	28.3
Firm Right	0.9	28.2